

Titanic Survival Prediction Using a Decision Tree Classifier

This project builds an end-to-end Decision Tree classification pipeline to predict passenger survival on the Titanic, chosen deliberately for the algorithm's defining strength: every prediction is a readable chain of if/else questions, making the fitted tree itself the deliverable. The dataset comprises 891 passengers described by 12 columns, of whom 38.4% survived (342 of 549 perished), with realistic data-quality problems handled first: 177 missing Age values were median-filled, 2 missing Embarked values mode-filled, and the Cabin column dropped at 77% missing.

Categorical features were converted with intent — Sex mapped to a binary 0/1 and Embarked one-hot encoded with the reference category dropped to avoid redundancy — yielding eight predictive features: passenger class, sex, age, siblings/spouses aboard, parents/children aboard, fare, and two embarkation indicators. After an 80/20 split (712 training, 179 test passengers), a DecisionTreeClassifier was trained using Gini impurity with maximum depth deliberately capped at 4, producing a compact 15-leaf tree — the central modelling decision, since an unconstrained tree memorises training data while a shallow one generalises.

On the 179-passenger test set the model achieved 79.9% accuracy against a majority-class baseline of 58.7% — a 21.2-point lift — with 0.839 precision and 0.635 recall for the survivor class (confusion matrix: 96 true negatives, 9 false positives, 27 false negatives, 47 true positives). The precision–recall profile is read honestly: when the model predicts survival it is right 84% of the time, but the conservative shallow tree misses roughly a third of actual survivors.

The most compelling output is what the tree learned. Feature importances concentrate 78% of predictive power in just two variables — Sex (58.0%) and passenger class (20.0%), with Fare (8.1%) and Age (7.6%) trailing — and the tree's structure reads as a data-derived retelling of the evacuation's social rules: the root split is Sex; for male passengers the next question is whether age is at most 6.5 years (the “children first” branch), while for female passengers it is cabin class, capturing the privilege gradient from first to third class.

The project closes with a full `plot_tree` visualisation — Gini values, sample counts, and class outcomes at every node — delivering a model whose reasoning any stakeholder can trace by hand, and demonstrating why interpretable models remain the right first choice when explanation matters as much as accuracy.